

Assume that  $C = A - I_3$  where

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -2 & 0 \\ -1 & 0 & 0 \end{bmatrix}.$$

Find the matrix  $B$  such that

$$CB^T = 3 \begin{bmatrix} -1 & 2 \\ 3 & 1 \\ 0 & -1 \end{bmatrix}$$

where  $I_3$  is the 3 by 3 identity matrix.

- ☐  $B = \begin{bmatrix} -3 & 4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$
- ☐  $B = \begin{bmatrix} -3 & -4 & 3 \\ 9 & 2 & 6 \end{bmatrix}$
- ☐  $B = \begin{bmatrix} -3 & -4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$
- ☐ None of the given options
- ☒  $B = \begin{bmatrix} 3 & -4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$

[Clear my choice](#)